QP Code:25901912

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU- UGP (Honours)/MGU- BBA (Honours)/MGU- BCA (Honours)
Special Examinations, July 2026
(for 2024 Admission Regular Examinations, October 2025)
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCMAT201
Building Blocks for Higher Mathematics
(2024 ADMISSION ONWARDS)
Scheme For Valuation

Duration: 2 Hours Maximum Marks: 70

Part A {Short Answer Questions}
Answer any 5 questions. Each question carries 2 marks

1. Explain the method of proof by contradiction. ... **[K] / [CO1]**

   - In the method of proof by contradiction, we assume that the statement to be proved is false. If this assumption leads to a contradiction (an impossible or false conclusion), then the assumption is wrong. Hence, the original statement must be true.

   ... 2 Marks

2. Find the number of relations from $A = \{a, b, c\}$ to $B = \{1, 2\}$. ... **[U] / [CO2]**

   - $|A \times B| = 3 \times 2 = 6$. ... 1 Mark
   - The number of relations is $2^6 = 64$. ... 1 Mark

3. Check if $\{-3, -1, 1, 3\}$ and $\{-2, 0, 2\}$ form a partition of the set $\{-3, -2, -1, 0, 1, 2, 3\}$. ... **[U] / [CO3]**

   - $$A \cap B = \emptyset$$

     and

     $$A \cup B = \{-3, -2, -1, 0, 1, 2, 3\}.$$

     Since the subsets are non-empty, mutually disjoint, and their union equals the given set, they form a partition.

     ... 2 Marsk

4. Give an example of an infinite lattice which has neither a least nor a greatest element. ... **[U] / [CO3]**

   - Any correct example like, The set of integers

     $$(\mathbb{Z}, \leq)$$

forms an infinite lattice, where

$$a \wedge b = \min(a, b), \qquad a \vee b = \max(a, b).$$

It has neither a least element nor a greatest element. ... 2 Marks

5. Find the unit tangent vector of the plane curve

$$\mathbf{r}(t) = \cos^3 t\,\mathbf{i} + \sin^3 t\,\mathbf{j}.$$

... [A] / [CO4]

- Calculations to find $= 3|\sin t \cos t|$. ... 1 Mark
- $\mathbf{T}(t) = \dfrac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = -\cos t\,\mathbf{i} + \sin t\,\mathbf{j}$, ... 1 Mark

6. Write a vector equation for the unit circle in the XY-plane centred at the origin. ... [A] / [CO4]

- A vector equation of the unit circle is

$$\mathbf{r}(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{j}, \qquad 0 \le t \le 2\pi.$$

... 2 Marks

7. Define the line integral of f over a curve C. ... [K] / [CO5]

- If C is a smooth curve parameterized by

$$\mathbf{r}(t), \quad a \le t \le b,$$

then the line integral of the scalar function f over C is

$$\int_C f\,ds = \int_a^b f(\mathbf{r}(t))\,|\mathbf{r}'(t)|\,dt.$$

... 2 Marks

8. Test whether the vector field

$$\mathbf{F} = (z + y)\mathbf{i} + z\mathbf{j} + (y + x)\mathbf{k}$$

is conservative. ... [U] / [CO5]

- Calculation of $\nabla \times \mathbf{F} = (0, 0, -1)$ ... 2 Marks
- $$\nabla \times \mathbf{F} = (0, 0, -1) \neq \mathbf{0}.$$

  Therefore,

  $\mathbf{F}$ is not conservative.

... 1 Mark

(Maximum Marks= 2)

$[\mathbf{2 \times 5 = 10}]$

Part B - Short Essay Questions
Answer any 5 questions. Each question carries 6 marks

9. (a) What do you mean by "begging the question"? ... **[A]/[CO1]**

- Begging the question is a logical fallacy in which the conclusion of an argument is assumed in one of its premises instead of being proved. Thus, the argument is circular and provides no independent evidence for the conclusion. ... 2 Marks

(b) Is the following argument correct? It supposedly shows that n is an even integer whenever n^2 is an even integer.

Suppose that n^2 is even. Then $n^2 = 2k$ for some integer k. Let $n = 2l$ for some integer l. This shows that n is even.

- No, the argument is **incorrect**. The statement $n = 2l$ is assumed without justification. This is precisely what needs to be proved. Hence, the proof commits the fallacy of *begging the question* ... 4 Marks

10. (a) What is meant by a direct proof? ... **[A]/[CO1]**

- A *direct proof* is a method of proving a statement by assuming that the given hypothesis is true and using definitions, axioms, and previously established results to logically deduce the conclusion. ... 2 Marks

(b) Give a direct proof to show that the sum of two even integers is even.

- $$a = 2m \quad \text{and} \quad b = 2n.$$

Therefore,

$$a + b = 2m + 2n = 2(m + n).$$

Since $m + n$ is an integer, let $k = m + n$. Then

$$a + b = 2k,$$

where k is an integer. Hence, $a + b$ is even.
Therefore, the sum of two even integers is even. ... 4 Marks

11. Write all the elements in $S \times S$ in lexicographic order based on the "less than" relation, where $S = \{1, 2, 3, 4\}$. ... **[U]/[CO3]**

- In lexicographic order, $(a, b) < (c, d)$ if either $a < c$, or $a = c$ and $b < d$. ... 2 Marks
- The elements of $S \times S$ in lexicographic order are

$$\{(1,1), (1,2), (1,3), (1,4),$$
$$(2,1), (2,2), (2,3), (2,4),$$
$$(3,1), (3,2), (3,3), (3,4),$$
$$(4,1), (4,2), (4,3), (4,4)\}.$$

... 4 Marks

12. Determine all minimal, maximal, least, and greatest elements of the poset $(\{1, 2, 3, 4, 6\}, |)$. ... **[U]/[CO3]**

- The partial order is given by divisibility ($|$), where $a \leq b$ if $a \mid b$. The divisibility relations are:

$$1 \mid 2,\ 1 \mid 3,\ 1 \mid 4,\ 1 \mid 6,\ 2 \mid 4,\ 2 \mid 6,\ 3 \mid 6.$$

... 2 Marks

- – **Minimal element** $\{1\}$,
  – **Maximal elements):** $\{4, 6\}$
  – **Least element:** 1
  – **Greatest element:** Does not exist

... 4 Marks

13. Find the arc length parameter $s(t)$ along the curve ... **[A] / [CO4]**

$$\mathbf{r}(t) = (4\cos t)\mathbf{i} + (4\sin t)\mathbf{j} + 3t\mathbf{k}$$

from the point where $t = 0$ by evaluating the integral

$$s = \int_0^t \|\mathbf{v}(\tau)\|\, d\tau.$$

Use this arc length parameter to find the length of the curve for

$$0 \leq t \leq \frac{\pi}{2}.$$

- Calculations to find Arc length parameter: $s(t) = 5t$ ... 3 Marks
- Calculations to find Length of the curve on $0 \leq t \leq \frac{\pi}{2} = \frac{5\pi}{2}$ ... 3 Marks

14. Find the unit tangent vector $\mathbf{T}$ and curvature of the helix ... **[A] / [CO4]**

$$\mathbf{r}(t) = (3\cos t)\mathbf{i} + (3\sin t)\mathbf{j} + 4t\mathbf{k}.$$

- Calculations to find the unit tangent vector,

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} = \frac{1}{5}\left(-3\sin t\,\mathbf{i} + 3\cos t\,\mathbf{j} + 4\,\mathbf{k}\right).$$

... 3 Marks

- Calculations to find the curvature,

$$\kappa = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|} = \frac{\frac{3}{5}}{5} = \frac{3}{25}.$$

... 3 Marks

15. Evaluate

$$\int_{(1,1,1)}^{(2,3,-1)} y\,dx + x\,dy + 4\,dz.$$

. . . **[A] / [CO5]**

- Calculations with proper results to find

$$\int_{(1,1,1)}^{(2,3,-1)} y\,dx + x\,dy + 4\,dz = 2 - 5 = -3.$$

. . . 6 Marks

16. A fluid's velocity field is

$$\vec{F} = (x - y)\mathbf{i} + x\mathbf{j}.$$

Find the circulation along the curve

$$\vec{r}(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{k}, \qquad 0 \le t \le 2\pi.$$

. . . **[An] /[CO5]**

- Calculations to get $\vec{F} \cdot d\vec{r} = -\sin t \cos t$. . . . 2 Marks
- Calculations to get

$$\begin{aligned}\int_C \vec{F} \cdot d\vec{r} &= \int_0^{2\pi} -\sin t \cos t\,dt \\ &= -\frac{1}{2}\int_0^{2\pi} \sin 2t\,dt \\ &= \frac{1}{4}\Big[\cos 2t\Big]_0^{2\pi} \\ &= \frac{1}{4}(1 - 1) = 0.\end{aligned}$$

. . . 4 Marks

[6 × 5 = 30]

Part C - Essay Questions
Answer any 3 questions Each question carries 10 marks

17. Determine whether the relations represented by the matrices are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive. . . . **[A]/[CO3]**

- $$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Property	Result
Reflexive	Yes
Irreflexive	No
Symmetric	Yes
Antisymmetric	No
Transitive	Yes

. . . 3 Marks

- $$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Property	Result
Reflexive	No
Irreflexive	No
Symmetric	No
Antisymmetric	Yes
Transitive	Yes

. . . 3 Marks

- $$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

. . . 3 Marks

Property	Result
Reflexive	No
Irreflexive	No
Symmetric	Yes
Antisymmetric	No
Transitive	No

- $$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

. . . 3 Marks

Property	Result
Reflexive	No
Irreflexive	No
Symmetric	No
Antisymmetric	No
Transitive	No

Maximum Marks =10

18. Prove that every non-empty set of natural numbers, ordered by the usual relation $\leq$, is well-ordered. . . . **[U]/[CO3]**

    - Correct Proof . . . 10 Marks

19. Let R be the set of all real numbers. Define a relation $\mathcal{R}$ by

$$x \,\mathcal{R}\, y \iff |x - y| > 0.$$

    Check all properties of an equivalence relation and determine whether $\mathcal{R}$ is an equivalence relation. . . . **[A]/[CO3]**

- Arguments for the relation is **not reflexive**. ...3 Marks
- Arguments for the relation **Symmetric:** ...3 Marks
- Arguments for the relation is not **Transitive:** ...3 Marks

Since $\mathcal{R}$ is not reflexive and not transitive, it is not an equivalence relation. ...1 Mark

20. Find the derivative of the function at P_0 in the direction of $\mathbf{u}$.

- Calculations upto

$$\begin{aligned} D_{\mathbf{u}}f &= \nabla f \cdot \widehat{\mathbf{u}} \\ &= (2,4,-6) \cdot \frac{1}{\sqrt{3}}(1,1,1) \\ &= \frac{2+4-6}{\sqrt{3}} = 0. \end{aligned}$$

...5 Marks

- Calculations upto

$$\begin{aligned} D_{\mathbf{u}}g &= \nabla g \cdot \widehat{\mathbf{u}} \\ &= (3,0,0) \cdot \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right) \\ &= 2. \end{aligned}$$

...5 Marks

21. Evaluate the line integral of the vector field

$$\mathbf{F}(x,y,z) = (y+z)\mathbf{i} + (z+x)\mathbf{j} + (x+y)\mathbf{k}$$

along the path $C_1 \cup C_2$ consisting of the line segment from $(0,0,0)$ to $(1,1,0)$ followed by the segment from $(1,1,0)$ to $(1,1,1)$. ...**[A]/[CO5]**

- The line integral is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r}.$$

...3 Marks

- Calculations to get

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_0^1 2t\, dt = \left[t^2\right]_0^1 = 1.$$

...3 Marks

- Calculations to get

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_0^1 2\, dt = 2.$$

...3 Marks

- $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r} = 3$ ...1 Mark

22. Find the divergence of each of the following vector fields and interpret its physical meaning. ...**[A]/[CO5]**

- **Expansion:**

$$\mathbf{F}(x, y, z) = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}.$$

Solution:

The divergence is

$$\nabla \cdot \mathbf{F} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3.$$

. . . 3 Marks

Physical Interpretation:

Since the divergence is positive, the field acts as a **source**. Fluid flows outward from every point, indicating uniform expansion. . . . 2 Marks

- **Compression:**

$$\mathbf{F}(x, y, z) = -x\hat{\mathbf{i}} - y\hat{\mathbf{j}} - z\hat{\mathbf{k}}.$$

Solution:

The divergence is

$$\nabla \cdot \mathbf{F} = \frac{\partial(-x)}{\partial x} + \frac{\partial(-y)}{\partial y} + \frac{\partial(-z)}{\partial z} = -1 - 1 - 1 = -3.$$

. . . 3 Marks

Physical Interpretation:

Since the divergence is negative, the field behaves as a **sink**. Fluid converges toward every point, indicating uniform compression. . . . 2 Marks

- **Rotation about the z-axis:**

$$\mathbf{F}(x, y, z) = -y\hat{\mathbf{i}} + x\hat{\mathbf{j}}.$$

Solution:

The divergence is

$$\nabla \cdot \mathbf{F} = \frac{\partial(-y)}{\partial x} + \frac{\partial x}{\partial y} + \frac{\partial 0}{\partial z} = 0 + 0 + 0 = 0.$$

. . . 3 Marks

Physical Interpretation:

Zero divergence means there is **no net source or sink**. The field represents pure rotation about the z-axis, with no local expansion or compression of the fluid. . . . 2 Marks

Maximum Marks=10